Research Plan

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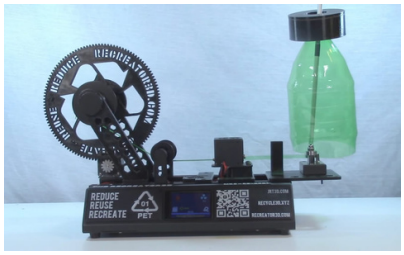
Designs are may differ from theoretical models presented due to the practical nature of the research.

Rationale

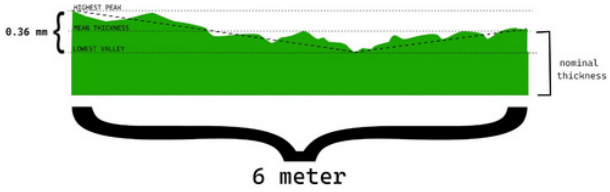
Single-use PET bottles, while providing numerous applications in our world, are a major source of microplastic contamination and hence proper recycling of such elements is crucial to achieving sustainable environmental goals. While there are several practices prevalent that partially serve this agenda such as recycling of PET bottles into fibers for cloth production, they fall short when it comes to the long-term sustainability of the final product (polyester clothing) and are attributed to "Downgrading" which renders the PET unrecyclable after conversion to fibers.



Standard PET recycling pultrusion system

PET filament pultrusion technique is also a technique for recycling PET bottles however it aims to fix the problem with a more liberal approach. By producing 3D printing filament with PET, users have the ability to generate any item/part based on their specific requirements with that item/part serving value for an indefinite period of time depending on its usage, maintenance, etc. However, certain limitations plague the overall productiveness of PET filament pultrusion.

The plastic bottles obtained from landfills (due to the constant stresses imparted on them such as temperature, pressure etc. for extended periods of time) exhibit degradation(and/or fragmentation) and hence when converted into strips, the strips do not possess a consistent thickness (volume) throughout their lengths which also reflects in the quality of the filament produced by them.

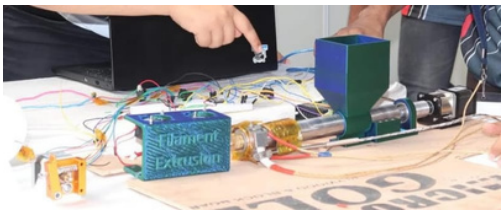


Simplified side profile of the strip

Hence this is where we have put forward our research into, increasing the effectiveness and overall versatility of PET filament pultrusion process while still maintaining its affordability..

The development of a technique aimed at enabling the pultrusion system to handle heavily degraded or damaged bottles is anticipated to yield the subsequent benefits:

1. Incentivising individual users, schools and print farms possessing access to 3D printers to construct and deploy the pultrusion system within their residences for filament production. This initiative would consequently mitigate energy consumption and reduce the carbon footprint associated with transporting discarded bottles to recycling facilities.
2. Offering end users the capability to manufacture filament from PET bottles would diminish reliance on commercial filaments, thereby resulting in a displacement of their usage. This shift holds the potential to contribute positively towards environmental sustainability.
3. Compared to small scale custom built extrusion systems that also produce 3D printing filament but via extrusion technique- with an all-metal camber as well as auger mechanism with relatively larger heating elements- the pultrusion system is constructed out of mainly spare components off of a 3d printer and 3d printed parts. Hence, the pultrusion system is less complex and more affordable in terms of overall construction and maintenance compared to equally sized extrusion systems for small scale deployment.



Small scale filament extrusion system

Research Questions

Before starting our research, we laid out clear foundational questions that we would try to abide by or answer to with the help of our research, these questions include:

1) What are some approaches for combating the problem of volumetrically inconsistent filaments?

2) Are current solutions commercially viable? If not then how to make them more generalized?

3) What is the cost of existing solutions? Are they affordable enough to be implemented in the PET pultrusion mechanisms commercially?

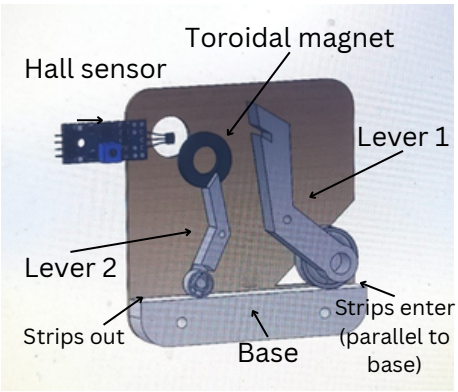
4) What can we design or add to previous solutions that would aid the process overall?

5) How to make it affordable?

6) Can the solution be open source, so that virtually anyone can implement it?

Hypothesis

Based on an exploration of current market options for sensors addressing inconsistent volumetric profiles of filaments used in standard 3D printers, we identified two critical criteria for potential innovation: cost-effectiveness and scalability. Through iterative discussions and collaborations, we conceptualized a novel solution termed the "volumetric gradient sensor."



CAD of prototypic volumetric gradient sensor

This proposed sensor design utilizes a hall effect sensor to measure the thickness of filament strips in real-time, prior to their conversion into 3D printing filament. The sensor comprises two levers arranged sequentially.

The first lever (1) incorporates a heated (60 °C) and relatively heavy ball bearing at its lower end. Its function is to apply pressure on the strips, removing large surface artifacts like indentations or groves and ensuring their close contact with the sensor base. The second lever (2), positioned after the first, features a smaller ball bearing at its lower end and a magnet at the other. This lever is designed to amplify the magnet's movement in response to the varying profiles of the strips. The resulting magnetic field fluctuations are detected by the hall effect sensor and transmitted to a microcontroller.

In collaboration with our proprietary code, the microcontroller dynamically adjusts the pull rate of the filament. In regions with lower volumetric output, the pull rate decreases to compress the filament, while in regions with higher volume, the pull rate increases to stretch the filament. This coordinated control mechanism aims to maintain a consistent volumetric output of the filament during the printing process.

Engineering Goals

The engineering goals will mainly stem from pursing a design for our prototypic sensor which is not only practical and easy to produce but also affordable to keep it grounded with respect to the pultrusion system.

CAD models for both the volumetric gradient sensor and the testing rig (standard pultrusion system) have been designed prior to experimentation in order to prepare appropriate STL files for 3D printing parts. The engineering goals can be divided into two subcategories :

i) **Volumetric gradient sensor**: The Volumetric Gradient Sensor will feature a support structure crafted from balsa wood for enhanced structural rigidity compared to 3D-printed alternatives. Lever 1, incorporating a roller heated to 50 to 60 degrees Celsius, will be fabricated from ABS plastic or a similarly heat-resistant material. Precision holes will be tapped for pivot placement. While the base will initially be 3D-printed, consideration is being given to employing a metal base (preferably made from aluminium) for consistent strip heating.
The sensor will utilize a DRV5055 A4 hall effect sensor with a sensitivity of 12.5 mV/mT and a quiescent voltage of 5V, ensuring compatibility with most standard development boards. Displacement measurement in Lever 1 will be facilitated by an N52 neodymium magnet (20*5*5 mm). Lever 1 will incorporate a 608zz ball bearing with an outer diameter of 22 mm, while Lever 2 will feature a 685zz ball bearing with an outer diameter of 5 mm.

ii) **Generic pultrusion system :** Our engineering approach focuses on leveraging the simplicity and accessibility of open-source PET recycling pultrusion systems available online.To ensure compatibility with a wide range of PET pultrusion setups built upon this framework, we will strictly adhere to the provided blueprints.
This entails utilizing a wooden base plate as outlined in the online blueprints, facilitating the installation of additional components such as the puller mechanism, nozzle, and spiral cutter. The puller mechanism will be predominantly 3D printed, with the exception of 8mm stainless steel rods used for axles. The Nema 17 stepper motor used for powering the puller mechanism is also excellent for precise variable speed control.
Moreover, our designs emphasize the repurposing of standard 3D printer parts to streamline construction. Notably, we will utilize L support brackets designed for stepper motors to serve as supports for the pultrusion system's nozzle, enhancing both efficiency and cost-effectiveness.

Procedure

Collection and Sorting: we will collect PET bottles from recycling centers and sort them based on their color (clear, green, blue, etc.) and type (1 litre, 2 litre, etc.).

Removal of Labels and Caps: We will remove any labels, stickers, and caps from the bottles. This will be done manually for our purposes; however, automated label removers can be used for the process.

Washing and Cleaning: The bottles will be washed thoroughly to remove any residue or contaminants. A mild detergent and water solution will be used, and they will be rinsed properly to ensure cleanliness. Consideration will be given to using a washing machine specifically designed for PET bottle cleaning if processing large quantities.

Drying: After washing, the bottles will be allowed to dry completely. Due to the relative number of bottles to be tested, we will dry the bottles using the sun, but industrial dryers could be used for large scales.

Strip conversion: The bottles will be cut in a spiral fashion, forming 6-meter-long singular strips from each 2 litre bottle for testing.

Feeding: The strips will be placed in the PET pultrusion system, connected to the puller mechanism, and set up for conversion into 3D printing filament.

Processing via volumetric gradient sensor: The sensor will record the thickness of each segment of the strip as the strips are in motion and the information will be relayed to the microcontroller board.

Computational assessment: The data from the gradient sensor will be used to compute the appropriate instantaneous change in pull rate needed for the correction of volumetric gradients if present in the strips. This would be executed at the time that, that particular segment of the strip reaches the nozzle.

Volumetric gradient correction: As a segment of the strip reaches the nozzle- where it would be molded into 3D printing filament- the microcontroller variates the pull rate of the strips and hence either stretches or squeezes the segment by a calculated amount in accordance with the computational assessment.

Produced filament Storage: This is the final stage, where the produced filament will be collected by wounding around a geared spool in the puller mechanism.

Risk and Safety

The safety assessment for the following research can be broken down into two subcategories:

i) Physical risks: These risks include burn risks from the heated nozzle operating at 230-240 degrees Celsius. A 3D-printed casing will be designed for the heating block and nozzle to prevent accidental contact with the parts during operation.

ii) Chemical risks: This includes possible VOC emissions during the production or testing of the filament produced from the PET material. To mitigate this concern, it was decided that all testing and production of the filament would take place in an open environment with proper air circulation. Furthermore, an adequate sensor would be installed in the system to track the emissions of VOCs during operation and printing.

Data Analysis

Experimentation with various parameters such as the operating frequency of the Hall effect sensor, the temperature of the roller, and the nozzle will be conducted in order to maximize effectiveness.

After obtaining certain values for each variable by experimentation that would lead to some operation, other variables would be isolated before measuring the effect of each variable individually on the volumetric gradient reduction.
Once the system is optimized to the highest degree, the relation between the change in pull rate and specific volumetric correction for that given change in pull rate for a fixed nozzle & roller temperature, pull rate and frequency of hall sensor would be plotted.

The testing for the relation between the above-stated parameters and the respective volumetric correction would be performed as follows:

Procedure : The strips, after drying, would be cut out into segments each measuring 2 meters in length and 10 equally spaced spots will be marked on the sides of the segments. Following this, the strips will pass through the sensor where the thickness will be noted for the marked spots. The strips will then be processed in the pultrusion mechanism enabled with our active correction model. The diameter of the filament produced at the same marked spots will be measured again with a digital vernier calliper and contrasted with the expected diameter, depending upon the nozzle, to get the value for the deviation of its diameter from the nominal diameter at the marked segments.

The graph pertaining to the change in instantaneous pull rate vs the volumetric change/correction would be possibly the most important relation to be mapped as it would help us formulate a mathematical function correlating with the graph so as to easily input the thickness as one parameter and obtain the respective value for the change in pull rate for correction of the volumetric gradient as the output.

Expected Outcomes

The expected outcomes from this research include the following:

1) Design of a sensor that can mitigate previous concerns related to PET recycling pultrusion systems.

2) Prototype roughly 8 times more affordable than previous solutions.

3) Prototype consisting majorly of generic and common parts.

4) Reduction in volumetric gradients in the filament by over 50% compared to the respective strips used, to ensure low probability of under extrusion or clogs.

5) Product filament suitable for application in a 3D printer without any further complications.

Bibliography

1)https://www.academia.edu/45680042/Open_Source_3_D_Filament_Diameter_Sensor_for_Recycling_Winding_and_Additive
2)
https://www.sciencedirect.com/science/article/abs/pii/S0014305720315883
3)
https://www.researchgate.net/publication/290429725 Crystallization Behavior of PET Materials
4)
https://www.mdpi.com/2076-3417/11/16/7338